Evaluating the Performance of Large Language Models on GAOKAO Benchmark

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Abstract

Large Language Models(LLMs) have demonstrated remarkable performance across various natural language processing tasks; however, how to comprehensively and accurately assess their performance becomes an urgent issue to be addressed. This paper introduces GAOKAO-Bench, an intuitive benchmark that employs questions from the Chinese GAOKAO examination as test samples, including both subjective and objective questions. To align with human examination methods, we design a method based on zero-shot settings to evaluate the performance of LLMs. With human evaluation, we obtain the converted total score of LLMs, including GPT-4, ChatGPT and ERNIE-Bot. Our findings reveal that LLMs have achieved competitive scores in Chinese GAOKAO examination, while they exhibit significant performance disparities across various subjects. We also use LLMs to grade the subjective questions, and find that model scores achieve a moderate level of consistency with human scores. In conclusion, this research contributes a robust evaluation benchmark for future large language models and offers valuable insights into the advantages and limitations of such models. [1]

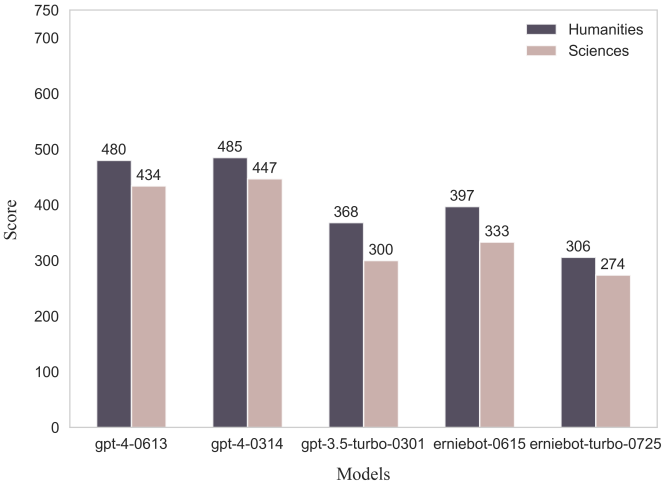

Figure 1: **Converted Total Score of LLMs in GAOKAO-Bench.** The converted total score for subjects in both the sciences and the humanities is 750 points.

1 Introduction

LLMs have demonstrated great abilities in handling diverse applications. The LLMs (Brown et al., 2020; Ouyang et al., 2022, OpenAI, 2023; Bubeck et al., 2023; Wei et al., 2022) indicate they possess abundant intrinsic knowledge, the ability to follow instructions and reasoning capabilities, which in certain areas are on par with or even

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[1]Resources are available through: https://github.com/OpenLMLab/GAOKAO-Bench

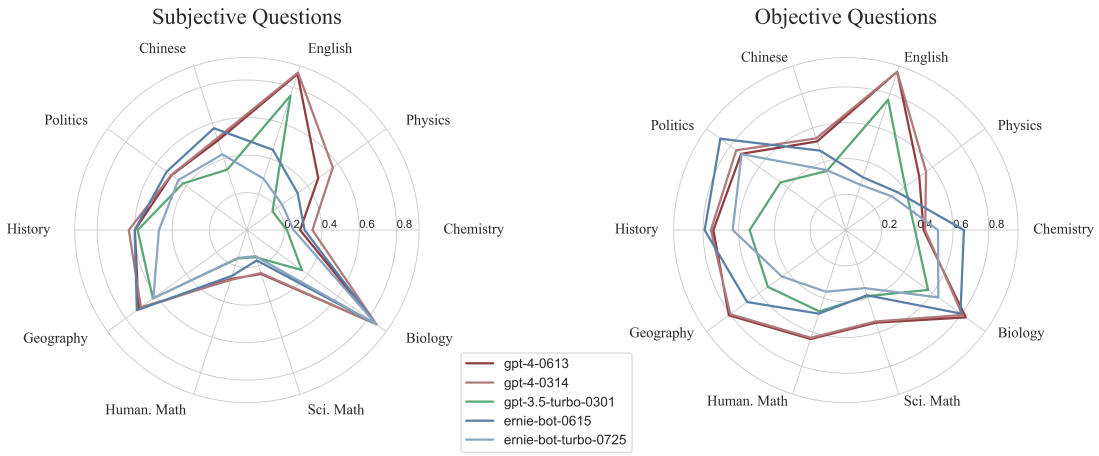

Figure 2: Scoring Rate of LLMs on objective and subjective questions across the subjects.

surpass human abilities. To better measure the capabilities of LLMs, researchers have proposed more comprehensive and challenging benchmarks. MMLU (Hendrycks et al., 2021) aims to measure a text model's multitask accuracy, covering 57 tasks such as elementary mathematics, US history, computer science, and more. BIG-Bench (Srivastava et al., 2022) introduces a comprehensive evaluation framework encompassing more than 204 subtasks, including linguistics, child development, among others. AGIEval (Zhong et al., 2023) evaluates the performance of LLMs in the context of human-centric standardized examinations and contains both Chinese and English tasks. Huang et al. (2023) propose C-Eval, a comprehensive Chinese evaluation suite covering four difficulty levels. However, the benchmark mentioned above only consists of objective questions and lacks subjective questions that are more closely related to generative abilities. Besides, due to the absence of real-world test samples, individuals often underestimate the complexity of these tasks and the abilities of the models, particularly in the context of the rapid development of LLMs. Consequently, there is a need for an intuitive and practical evaluation method.

We propose using the Chinese College Entrance Examination (GAOKAO) questions. These questions include computational, reasoning, knowledge assessment and writing tasks (Tan et al., 2021). Previous benchmarks based on the GAOKAO mainly focus on English (Yuan and Liu, 2022), especially English Reading and Comprehension Questions (Zhang et al., 2022). To this end, we introduce the GAOKAO-Benchmark (GAOKAO-Bench), a benchmark specifically tailored to LLMs evaluation that covers the GAOKAO questions from 2010 to 2022. The GAOKAO-Bench consists of 9 subjects with 1781 objective questions and 1030 subjective questions. The question types include single-choice, cloze, correction, open-ended questions, and more.

We conduct experiments on some currently best-performing LLMs. To more accurately measure their generative capabilities, we use human scoring evaluation to judge subjective questions. The results in Figure1 show that LLMs have achieved competitive scores in the GAOKAO. Meanwhile, we find that all of the LLMs exhibit obvious signs of subject bias, which informs the future development of LLMs.

Due to the high cost of human evaluation, we provide human-annotated marking criteria of subjective questions. And we use LLM as a judge to evaluate LLMs on subjective questions. The results indicate that equipped with the detailed marking criteria, LLMs exhibit high consistency with human teachers, making the large-scale assessment of subjective questions feasible.

2 GAOKAO-Bench

2.1 Introduction to the GAOKAO

The Chinese College Entrance Examination, also known as the GAOKAO, is a nationwide examination designed to assess the academic abilities of high school students applying to universities in China. Known as a rigorous and comprehensive examination, the GAOKAO is differentiated into two distinct streams: the sciences and the humanities: the sciences include Chinese, sciences mathematics, English, physics, chemistry and biol-

Models	Overall	Chinese	Eng.	Sci. Math	Hum. Math	Phys.	Chem.	Biol.	Poli.	Hist.	Geog.
LLaMA-7b	21.1%	16.2%	20.5%	24.3%	26.1%	0.0%	22.6%	22.7%	22.2%	19.2%	24.2%
Vicuna-7b	21.0%	12.0%	19.6%	23.8%	23.4%	7.0%	27.4%	20.0%	20.9%	23.0%	23.2%
Baichuan2-7b-Base	27.2%	16.2%	21.2%	24.8%	24.8%	0.0%	23.4%	24.0%	55.3%	32.1%	24.2%
Baichuan2-7b-Chat	40.5%	31.7%	33.0%	26.6%	28.4%	18.0%	26.6%	48.0%	69.7%	57.8%	49.5%
Baichuan2-13b-Chat	43.9%	26.9%	34.7%	23.8%	31.7%	25.0%	40.3%	53.3%	75.3%	59.9%	61.1%
ChatGLM-6b	30.8%	18.6%	17.0%	25.2%	25.7%	12.5%	30.6%	24.7%	54.1%	59.9%	25.3%
ChatGLM2-6b	42.7%	31.1%	30.6%	29.0%	35.8%	24.2%	46.0%	71.3%	55.0%	59.2%	41.1%
GPT-4-0613	71.6%	52.1%	**93.2**%	**54.5**%	**64.0**%	50.8%	43.6%	**83.0**%	72.5%	74.2%	**81.1**%
GPT-4-0314	**72.2**%	**53.9**%	93.1%	53.7%	63.3%	**55.5**%	44.4%	80.7%	75.9%	75.6%	80.0%
GPT-3.5-turbo-0301	53.2%	34.7%	76.6%	38.8%	47.8%	41.1%	38.7%	56.9%	45.3%	53.9%	54.0%
ERNIE-Bot-0615	56.6%	46.7%	31.0%	38.3%	49.1%	35.9%	**66.1**%	79.3%	**86.9**%	**79.1**%	68.4%
ERNIE-Bot-turbo-0725	45.6%	35.3%	26.6%	34.1%	36.2%	32.0%	51.6%	64.0%	72.2%	63.4%	44.2%

Table 1: **Scoring Rate of Objective Questions.** Models above the line are open-source LLMs; models below the line are closed-source LLMs.

ogy; the humanities include Chinese, humanities mathematics, English, politics, history and geography. The examination encompasses a variety of question types that include logical reasoning, computational analysis, knowledge-based quizzes and written expression among other aspects.

2.2 Dataset Description

The GAOKAO-Bench established in this paper includes the content of all national exams in the GAOKAO of all subjects from 2010 to 2022, providing an intuitive and human-aligned evaluation benchmark for LLMs.
We obtain the questions and transform them from PDF into JSON file format using a combination of automated scripting and manual annotation. Mathematical formulas within the questions were converted into LaTeX format. Appendix A.1 provides an example of a mathematical single-choice question.

The questions are divided into subjective and objective categories, depending on whether they require human scoring. In total, we select 2811 questions, including 1030 objective questions and 1781 objective questions. Table 3 provides a breakdown of the specific types of questions and the corresponding number of questions in each type. Multi-Question Choice refers to a format where a single question is followed by multiple sub-questions and Multi-Choice refers to a format where a single question corresponds to multiple correct answers.

3 Experiments

3.1 Methodology

Prompt Design In order to emulate the format in which humans partake in examinations, we utilize a zero-shot settings strategy (Ouyang et al., 2022) and create prompts tailored to different question types. The prompts not only require the model to complete the task, but also explicitly specify the format of the output as we contend that the intrinsic knowledge level of the model and its ability to follow instructions are equally important. The specific prompt examples we use are illustrated in Appendix A.1.

Models We evaluate several current best-performing LLMs that support both Chinese and English:

1. GPT-4: We test on 2 checkpoints: GPT-4-0613 and GPT-4-0314.

2. ChatGPT: We test on GPT-3.5-turbo-0301 checkpoint.

3. ERNIE-Bot: A Chinese LLM published by

Models	Overall	Chinese	Eng.	Sci. Math	Hum. Math	Phys.	Chem.	Biol.	Poli.	Hist.	Geog.
GPT-4-0613	50.8%	50.3%	87.6%	**24.6%**	27.5%	47.1%	28.5%	**85.6%**	49.9%	59.9%	71.5%
GPT-4-0314	**51.9%**	51.5%	**88.3%**	24.1%	**27.9%**	**56.7%**	**35.0%**	**85.6%**	50.0%	**63.1%**	70.0%
GPT-3.5-turbo-0301	35.8%	33.9%	75.4%	15.2%	15.9%	16.9%	21.4%	36.3%	42.3%	58.4%	62.1%
ERNIE-Bot-0615	48.4%	**57.1%**	45.0%	17.0%	25.6%	33.5%	30.8%	84.9%	**53.0%**	60.0%	**72.7%**
ERNIE-Bot-turbo-0725	39.2%	42.5%	28.8%	14.6%	15.6%	23.2%	25.0%	85.1%	45.3%	47.0%	61.8%

Table 2: **Scoring Rate of Subjective Questions.** The results are scored by human teachers.

Question Type		Number	Percentage
Objective	Single Choice	1418	50.5%
	Multi-Question Choice	273	9.7%
	Multi-Choice	64	2.3%
	Five out of Seven	26	0.9%
Subjective	Open-ended Question	786	28.0%
	Cloze	218	7.8%
	Correction	26	0.9%

Table 3: Distribution of Question Types.

Baidu. We test on ERNIE-Bot-0615 checkpoint.

4. ERNIE-Bot-turbo: We test on ERNIE-Bot-turbo-0725 checkpoint.

We set the sampling temperature to 0.3 in order to achieve a balance between stability and diversity.

Metric When evaluating objective and subjective questions separately, we use the scoring rate $R_{i,\text{obj}}$ and $R_{i,\text{subj}}$ for each subject i.
To evaluate the overall performance, we convert the scoring rates of subjective and objective questions into a total score S_{total}. We mimic the subjective question scores as $M_{i,\text{subj}}$ and objective question scores $M_{i,\text{obj}}$ for each subject i in the GAOKAO. The converted total score can be formulated as:

$$S_{\text{total},\,\mathcal{S}} = \sum_{i\in\mathcal{S}} \left(R_{i,\text{obj}} \cdot M_{i,\text{obj}} + R_{i,\text{subj}} \cdot M_{i,\text{subj}}\right),$$
$$S_{\text{total},\,\mathcal{H}} = \sum_{i\in\mathcal{H}} \left(R_{i,\text{obj}} \cdot M_{i,\text{obj}} + R_{i,\text{subj}} \cdot M_{i,\text{subj}}\right)$$

where $\mathcal{S}$ stands for the set of the sciences subjects, and $\mathcal{H}$ stands for the set of the humanities subjects. The total scores of sciences and humanities are both 750 points. Detailed total score for each subject is shown in Appendix B.

3.2 Objective Questions

Each item i in the GAOKAO-Bench comprises the question q_i, the standard answer a_i, the score s_i, the analysis n_i. For objective questions, the input includes the question q_i and the LLMs need to output (r_i, o_i), where r_i denotes the corresponding reasoning process and o_i denotes the outcome. Points are awarded only if the outcome o_i is consistent with the standard answer a_i. Following the technical report for OpenAI's GPT-4 (OpenAI, 2023), we score the objective questions using regular matching. In addition to the LLMs mentioned above, we evaluate several open-source LLMs on GAOKAO-Bench, including LLaMA (Touvron et al., 2023), Baichuan (Yang et al., 2023)and ChatGLM (Zeng et al., 2023).

3.3 Subjective Questions

The input and output formats of the subjective questions are similar to those of objective questions. During the grading process, evaluators take into account both the reasoning process r_i and the outcome o_i. We assess the subjective questions using human scoring, in order to more precisely evaluate the performance of LLMs. Each subjective question is evaluated by two teachers, and the average of these scores was adopted as the final score for that question.

3.4 LLM as a Judge

Due to the high cost of manual evaluation, it is a natural progression to consider the use of LLMs for grading subjective questions. To better align with the teachers, we solicit teachers to provide detailed marking criteria m_i, breaking down the answers into specific scoring points for each item i. We design prompts in zero-shot settings and utilize GPT-4-turbo (GPT-4-1106-preview) as a judge. For each input $(q_i, a_i, s_i, n_i, m_i, r_i, o_i)$, the LLM need to output (g_i, f_i), where g_i denotes the process of

grading and f_i denotes the final score. The sampling temperature is set to 0 to obtain deterministic scores. We calculate the converted total score and Spearman and Kendall-Tau correlations between predicted scores and human scores following Jain et al. (2023) and Zhong et al. (2022).

3.5 Results

Overall Performance Figure 1 shows the converted total score of LLMs on GAOKAO-Bench. GPT-4 achieves scores exceeding 400 points and ERNIE-Bot surpasses ChatGPT. Every LLM obtains higher scores in humanities than in sciences. In the GAOKAO, the sciences require more advanced logical reasoning and computational steps than the humanities; and the humanities require a greater amount of knowledge than the sciences. The result indicates the reasoning and calculation abilities of LLMs still need further improvement.

Performance on Objective Questions Table 1 reflects the performance of LLMs on objective questions in different subjects. Open-source models pre-trained on Chinese language data and aligned with human perform better in all subjects. And the performance of the models improves with the increase in their scale. For closed-source LLMs, GPT-4 maintains a lead in the majority of subjects, but ERNIE-Bot performs better in chemistry, politics and history.

Performance on Subjective Questions Table 2 indicates the human evaluation of subjective questions. GPT-4 obtains the highest scoring rate (51.9%) and ERNIE-Bot achieves a comparably close level (48.4%). GPT-4 and ChatGPT exhibit superior performance in English compared to Chinese, whereas ERNIE-Bot and ERNIE-Bot-turbo demonstrate the opposite trend, excelling more in Chinese than in English.

LLM as a Judge Table 4 shows the results of using GPT-4-turbo to grade subjective questions. The Question-level Spearman and Kendall-Tau correlations show a markedly strong positive correlation between model judging and human scoring.

4 Analysis

4.1 Difference in Subjects

We analyze the scoring rate of subjective questions and objective questions in different subjects of LLMs, and find that there are large differences in the ability of the model in different subjects both in objective questions and subjective questions.
GPT-4 excels in English, biology and geography with scoring rates greater than 70% both in subjective and objective questions. However, they demonstrate poor performance in mathematics and physics with scoring rates less than 40%. ERNIE-Bot performs better in biology, history, politics in subjective questions with scoring rates greater than 60%, but the scoring rate of mathematics is less than 30%.
We posit that the substantial disparities across subjects can be attributed to two primary factors: firstly, the distinct competencies evaluated by each subject, for instance, language comprehension and summarization abilities in Chinese and English, and logical reasoning and computational skills in mathematics and physics; secondly, aspects related to the training of the model, including the richness of the pre-training corpus and the inclinations towards human alignment.

4.2 Difference between Sujective and Objective Questions

For a given subject, the scoring rate of subjective questions is generally lower than that of objective questions. For example, the scoring rate of subjective mathematics questions of GPT-4 is significantly lower than that on subjective mathematics questions. We hypothesize that subjective mathematics questions distinctly require the application of correct formulas, as well as more extensive computational and reasoning steps, which poses a significant challenge for LLMs. And compared to objective questions, the subjective questions of humanities necessitate students' mastery of more precise knowledge points, as well as their abilities in induction, summarization and categorical organization.

4.3 Stable Annual Trends on the GAOKAO

We categorize the examination questions based on their respective years and compute the model's converted total scores from 2013 to 2022 in Figure 3. We observe that the converted total score of LLMs are stable across the last decade. It indicates a rela-

Models	Sciences		Humanities		ρ	τ
	Human	GPT-4-turbo	Human	GPT-4-turbo		
GPT-4-0613	434	428	480	523	0.854	0.710
GPT-4-0314	447	440	485	535	0.852	0.709
GPT-3.5-turbo-0301	300	314	368	408	0.845	0.710
ERNIE-Bot-0615	333	348	397	436	0.864	0.740
ERNIE-Bot-turbo-0725	274	262	306	328	0.825	0.685

Table 4: Converted total score, Question-level Spearman and Kendall-Tau correlations of Human and GPT-4-turbo grading methods.

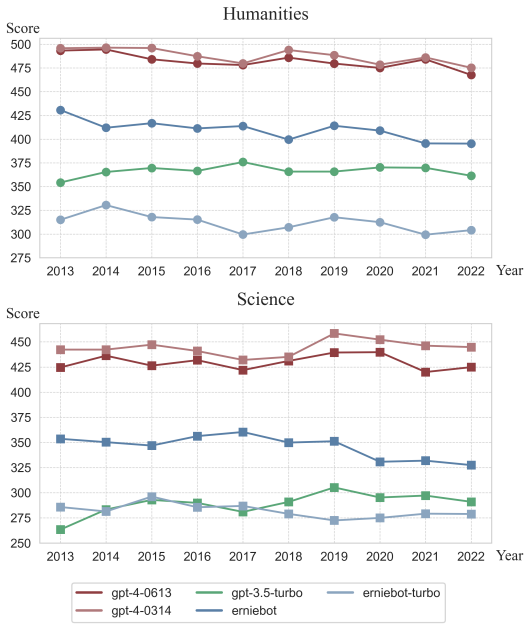

Figure 3: The Annual Trends of LLMs on GAOKAO-Bench.

tive stability in the difficulty level of the GAOKAO questions.

4.4 LLM as a Judge

We observe that the converted total score of sciences is much closer to human scoring than that of humanities. The deviation in scores for science subjects is less than 2% of the total score, and for humanities subjects, the deviation is around 5% of the total score. We posit that the answers and scoring criteria in the sciences are relatively explicit, whereas in the humanities, scoring depends on the alignment of semantics with designated points. This necessitates a fine-grained semantic understanding by the models, presenting a significant challenge for LLMs.

5 Avoid Benchmark Leakage

Benchmark leakage means the data related to evaluation sets is occasionally used for model training (Zhou et al., 2023). And it is plausible that the GAOKAO questions may be included in the training corpus of LLMs. The zero-shot settings and human evaluation used in this paper can alleviate the unfair phenomenon. Given that the GAOKAO is conducted annually in June, we plan to incorporate each year's new GAOKAO questions into the GAOKAO-Bench as a supplement, aiming to mitigate the issue of dataset leakage in evaluations. We have released the GAOKAO-Bench-2023 [2] which includes the objective questions in the 2023 GAOKAO. And we compare the scoring rate of objective questions in GAOKAO-Bench and GAOKAO-Bench-2023 in Table 5. We contend that these variations are within the normal range of difficulty fluctuations.

Models	GAOKAO-Bench	GAOKAO-Bench-2023	Δ
ChatGLM-6b	30.8%	24.1%	-6.7%
ChatGLM2-6b	42.7%	36.9%	-5.8%
Baichuan2-7b-chat	40.5%	37.9%	-2.6%
Baichuan2-13b-chat	43.9%	41.3%	-2.6%
GPT-4-0613	71.6%	71.0%	-0.6%
GPT-4-0314	72.2%	69.8%	-2.4%

Table 5: **Scoring Rate of Objective Questions on GAOKAO-Bench-2023.** The GAOKAO-Bench covers questions from 2010 to 2022.

6 Ablation Study

We investigate the impact of manually annotated marking criteria on the accuracy of the LLM's grading of subjective questions. We use the GPT-4-turbo to evaluate the performance of GPT-4, ChatGPT and ERNIE-Bot-turbo with or without marking criteria. Tabel 6 indicates that provided with

[2]Resources are available through: https://github.com/OpenLMLab/GAOKAO-Bench-2023

marking criteria, LLMs can better align with human preferences.

Methods		GPT-4-0613	GPT-3.5-turbo-0301	ERNIE-Bot-turbo-0725
w marking criterion	ρ	0.854	0.845	0.825
	τ	0.710	0.710	0.685
w/o marking criterion	ρ	0.820	0.820	0.803
	τ	0.659	0.674	0.654

Table 6: Spearman and Kendall-Tau Correlations of LLM grading and human judgement.

7 Related Work

Benchmark for LLMs The flourishing development of LLMs has also raised higher demands for benchmarks. Benchmarks for traditional tasks in NLP, such as GLUE (Wang et al., 2018) for natural language understanding, SQuAD (Rajpurkar et al., 2016) for reading comprehension, cannot measure the comprehensive capabilities of LLMs. Consequently, researchers have proposed new benchmarks to evaluate the advanced abilities of LLMs. MMLU (Hendrycks et al., 2021) provides a multi-task test across a diverse set of subjects. BIG-Bench (Srivastava et al., 2022) covers a diverse range of topics and languages, including auto debugging, know unknowns, logical deduction. HELM (Liang et al., 2023) taxonomies the design space of language model evaluation into scenarios and metrics. In the field of Chinese language benchmarks, C-Eval (Huang et al., 2023) selects multiple-choice questions across four difficulty levels: middle school, high school, college, and professional. AGIEval (Zhong et al., 2023) assesses LLMs in the context of human-centric standardized exams. CMMLU (Li et al., 2023) includes subjects that may not typically appear in standard exams but are relevant to people's daily life, such as Chinese food culture, Chinese driving rule.

Human evaluation for LLMs Compared to automatic evaluation, human evaluation is more aligned with real-world application scenarios and can offer more comprehensive and precise feedback (Chang et al., 2023). Chatbot Arena (Zheng et al., 2023) provides a platform to assess and compare diverse chatbot models through user engagement and voting. Ziems et al. (2023) adopts human scoring evaluation on generation tasks. Liang et al. (2023) conduct human evaluations on 6 LLMs on summarization and disinformation scenarios.

8 Limitations

While we evaluate and analyze the performance of LLMs on GAOKAO-Bench, there are some limitations in this work. Firstly, due to the constraints in time and resources, this paper does not delve into the detailed analysis of the errors made by LLMs on the GAOKAO-Bench, such as model hallucinations and reasoning mistakes. Secondly, due to the rapid developments of LLMs and high cost of human evaluation, we are unable to conduct experiments on every model using human scoring. We hope to enhance the evaluation and analysis of the models' reasoning process and utilize LLMs as a replacement for human scoring in future work.

9 Conclusion

In this paper, we introduce the GAOKAO-Bench dataset, which serves as an evaluation standard for large language models. The dataset includes Chinese College Entrance Examination questions from 2010 to 2022, covering various subjects and question types, with an overall high level of difficulty. By testing large language models on the GAOKAO-Bench, we can analyze the gap and advantages of these models compared to humans in a reasonable and intuitive manner.

In addition, we evaluate the ability of large language models to answer Chinese College Entrance Examination questions using zero-shot prediction approach and human evaluation. Our results show that the models perform well on knowledge-based questions, but struggle with certain types of logical reasoning and mathematical problems, as well as with reading comprehension of longer texts in Chinese.

We also use the LLMs to evaluate subjective questions, which is called LLM-as-a-Judge. We observe that equipped with human-annotated marking criteria, the LLM evaluation is consistent to human preference.

These findings suggest that large language models have potential applications in education and language assessment, but there is still room for improvement in certain areas. Future work could focus on developing approaches to enhance the model's performance on longer text reading comprehension tasks, logical reasoning and calculation problems.

Author Contributions

Qixiang Wang contributed to data collection and curation for GAOKAO-Bench.

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A Examples

A.1 Data Example

An example of the single-choice questions in sciences mathematics is shown below. The English is added by the author for readers' understanding.

- **Question**

若$z=-1+\sqrt{3}$i, 则$\frac{z}{z\bar{z}-1}=$(　　)
A. $-1+\sqrt{3}$i
B. $-1-\sqrt{3}i$
C. $-\frac{1}{3}+\frac{\sqrt{3}}{3}$i
D. $-\frac{1}{3}-\frac{\sqrt{3}}{3}i$

If $z=-1+\sqrt{3}$i, then $\frac{z}{z\bar{z}-1}=$(　　)
A. $-1+\sqrt{3}$i
B. $-1-\sqrt{3}i$
C. $-\frac{1}{3}+\frac{\sqrt{3}}{3}$i
D. $-\frac{1}{3}-\frac{\sqrt{3}}{3}i$

- **Answer**

C

- **Explanation**

【详解】
$\bar{z}=-1-\sqrt{3}i, z\bar{z}=(-1+\sqrt{3}i)(-1-\sqrt{3}i)=1+3=4.$
$\frac{z}{z\bar{z}-1}=\frac{-1+\sqrt{3}\text{i}}{3}=-\frac{1}{3}+\frac{\sqrt{3}}{3}$i
故选: C

The detailed solution for the given problem is as follows:
$\bar{z}=-1-\sqrt{3}i, z\bar{z}=(-1+\sqrt{3}i)(-1-\sqrt{3}i)=1+3=4.$
$\frac{z}{z\bar{z}-1}=\frac{-1+\sqrt{3}\text{i}}{3}=-\frac{1}{3}+\frac{\sqrt{3}}{3}$i
Therefore, the correct option is C.

A.2 Answer Example

An example of the answer of a geography open-ended question given by ChatGPT is shown below.

- **Prompt & Question**

请解答下面的地理解答题。仔细阅读题目并充分结合你已有的知识，解答其中的问题，请你一步步思考并将思考过程写在【解析】和<eoe>之间。你的答案请写在【答案】和<eoa>之间。
完整的题目回答格式如下:
(1)【解析】...<eoe>【答案】...<eoa>
(2)【解析】...<eoe>【答案】...<eoa>
请你严格按照上述格式作答，如果不止一道题，请分别作答。
题目如下:

位于陕西省安康市汉阴县的凤堰古梯田，是目前已发现的秦巴山区面积最大、保存最完整的清代梯田。在总面积约800 公顷的古梯田保护区内，分布有自清乾隆年间以来修建的古建筑民居群落、古寨堡、古庙宇、古堰渠、

古塘坝等文物遗存。2019 年，凤堰古梯田被列为全国重点文物保护单位。以古梯田为“展品”，以秀美的自然风光为“展厅”，凤堰古梯田被打造成保护和展示原生态生活方式、促进村镇科学发展的开放式“生态博物馆”。提出该开放式“生态博物馆”开展旅游活动的旅游资源保护措施。

Please answer the following geography essay question. Read the question carefully and fully integrate your existing knowledge to answer the questions. Think step by step and write your thought process in the section between 【Analysis】and <eoe>. Write your answer in the section between 【Answer】and <eoa>.
The complete format for answering the question is as follows:
(1)【Analysis】...<eoe>【Answer】...<eoa>
(2)【Analysis】...<eoe>【Answer】...<eoa>
Please strictly follow the above format to answer. If there is more than one question, please answer them separately.
The question is as follows:

The Fengyan Ancient Terraces, located in Hanyin County, Ankang City, Shaanxi Province, are the largest and most well-preserved Qing Dynasty terraces discovered in the Qinba Mountain area to date. Within the approximately 800-hectare ancient terrace conservation area, there are distributed cultural relics such as ancient residential buildings, ancient fortresses, ancient temples, ancient canals, and ancient dams, which have been constructed since the Qing Dynasty's Qianlong era. In 2019, the Fengyan Ancient Terraces were listed as a national key cultural relics protection unit. With the ancient terraces as the 'exhibit' and the beautiful natural scenery as the 'exhibition hall', the Fengyan Ancient Terraces have been transformed into an open 'Ecological Museum' that protects and displays the original ecological lifestyle and promotes the scientific development of villages and towns. Propose tourism resource conservation measures for conducting tourism activities in this open 'Ecological Museum'

- **Output of ChatGPT**

【解析】
凤堰古梯田是一处重要的文化遗产，被列为全国重点文物保护单位。为了保护这一遗产，同时也为了开展旅游活动，需要采取一些措施来保护旅游资源。
【答案】
1. 加强管理：对凤堰古梯田保护区内的游客进行管理，设置必要的警示标志，禁止破坏梯田、采摘植物等行为。同时，加强对古建筑民居群落、古寨堡、古庙宇、古堰渠、古塘坝等文物遗存的保护，防止游客在参观过程中对这些文物遗存造成损害。
2. 推广科普：在凤堰古梯田保护区内设置科普展板，向游客介绍梯田的历史、文化和生态环境，提高游客的文化素养和环保意识，减少游客对梯田的破坏。
3. 限流措施：为了保护梯田的生态环境，可以设置游客限流措施，限制游客进入梯田的范围和数量，减少游客对梯田的踩踏和破坏。
4. 建设配套设施：为了方便游客参观，可以在凤堰古梯田保护区内建设一些配套设施，如停车场、休息区、卫生间等，提高游客的参观体验。同时，这些设施也需要合理规划，避免对梯田的破坏。

【Analysis】
The Fengyan Ancient Terraces are an important cultural heritage and have been listed as a national key cultural relics protection unit. In order to protect this heritage while also developing tourism activities, measures need to be taken to protect the tourism resources.
【Answer】
1. Strengthen Management: Manage the tourists within the Fengyan Ancient Terraces conservation area, set up necessary warning signs, and prohibit actions such as damaging the terraces or picking plants. At the same time, enhance the protection of cultural relics such as ancient residential buildings, ancient fortresses, ancient temples, ancient canals, and ancient dams

to prevent visitors from damaging these relics during their visit.
2. Promote Science Education: Install educational boards within the Fengyan Ancient Terraces conservation area to introduce tourists to the history, culture, and ecological environment of the terraces. This will enhance the cultural literacy and environmental awareness of the tourists and reduce the damage to the terraces.
3. Implement Visitor Limitations: To protect the ecological environment of the terraces, visitor limitations can be set to restrict the range and number of tourists entering the terraces, thus reducing trampling and damage.
4. Develop Supporting Facilities: To facilitate visitor access, some supporting facilities such as parking lots, rest areas, and restrooms can be constructed within the Fengyan Ancient Terraces conservation area, enhancing the visitor experience. At the same time, these facilities should be planned judiciously to avoid damage to the terraces.

A.3 Model Grading Example

We show the grading process of GPT-4-turbo corresponding to the answer given by ChatGPT in Appendix A.2.

- **Prompt**

你是一名高中地理老师，正在批改高考地理试卷。
请根据下面的【题目】、【分析过程】、【标准答案】、【分值】、【评分点】、【学生分析与答案】，对【学生分析与答案】进行判分并给出理由。
请注意【学生分析与答案】可能为空。
如果有多道小题，分别给出每道小题的得分，并在最后给出最终得分。
输出格式为：【判分理由】...【得分】... ...【总分】...分。
其中【总分】直接给出这道题的最终分数，如【总分】5分，注意不要超过这道题的【分值】。
请严格对照【标准答案】和【评分点】中的得分点。

You are a high school geography teacher currently grading a geography paper from the college entrance examination.
Please score the 'Student Analysis and Answer' based on the following sections: 'Question', 'Analysis Process', 'Standard Answer', 'Points', 'Scoring Criterion', and 'Student Analysis and Answer'.
Note that the 'Student Analysis and Answer' section might be empty. If there are multiple sub-questions, provide scores for each sub-question separately and give the total score at the end.
The output format should be: 【Scoring Rationale】...【Points Awarded】...【Total Points】... points.
The 【Total Points】should directly present the final score for the question, for example, 【Total Points】5 points. Be careful not to exceed the total points available for the question.
Please strictly compare against the 'Standard Answer' and the scoring points in the 'Scoring Criterion'.

- **Marking Criterion**

【答案】完善景区规划、依法保护生态环境，对古建筑等进行保护性修缮，尽可能保证新建设施与景区景观相融合。倡导绿色旅游，普及旅游文化环境保护教育，提高游客对旅游资源环境保护的意识。制定景区相关环境保护条例与惩罚制度，评定该"生态博物馆"的环境容量，对人口数量的容纳程度，限制客流量。（任答三点10分）

【Answer】Improve the planning of the scenic area, protect the ecological environment in accordance with the law, and carry out conservation repairs on ancient buildings to ensure that new facilities blend harmoniously with the scenic landscape. Advocate for green tourism, popularize education on the protection of the tourism cultural environment, and raise tourists' awareness of the protection of tourism resources and environment. Develop environmental protection regulations and punishment systems for the scenic area, assess the environmental capacity of this

‘Ecological Museum’, and regulate the carrying capacity in terms of population, limiting the flow of visitors. (Any three points for 10 points.)

- **Output of GPT-4-turbo**

【判分理由】
1. 加强管理：学生提出了对游客进行管理和设置警示标志，以及加强对文物遗存的保护，这与标准答案中的“完善景区规划、依法保护生态环境”和“制定景区相关环境保护条例与惩罚制度”相符，可以给分。【得分】3分
2. 推广科普：学生提出了设置科普展板来提高游客的文化素养和环保意识，这与标准答案中的“普及旅游文化环境保护教育，提高游客对旅游资源环境保护的意识”相符，可以给分。【得分】3分
3. 限流措施：学生提出了设置游客限流措施，这与标准答案中的“评定该‘生态博物馆’的环境容量，对人口数量的容纳程度，限制客流量”相符，可以给分。【得分】2分
4. 建设配套设施：学生提出了建设配套设施并合理规划，这与标准答案中的“尽可能保证新建设施与景区景观相融合”相符，可以给分。【得分】2分
【总分】10分

【Scoring Rationale】
1. Strengthen Management: The student proposed managing tourists and setting up warning signs, as well as enhancing the protection of cultural relics, which aligns with the standard answer’s "improve the planning of the scenic area, protect the ecological environment in accordance with the law" and "develop environmental protection regulations and punishment systems for the scenic area." Points can be awarded for this. 【Points Awarded】3 points
2. Promote Science Education: The student proposed setting up educational boards to enhance tourists’ cultural literacy and environmental awareness, which aligns with the standard answer’s "popularize education on the protection of the tourism cultural environment, raise tourists’ awareness of the protection of tourism resources and environment." Points can be awarded for this. 【Points Awarded】3 points
3. Implement Visitor Limitations: The student proposed visitor limitation measures, which align with the standard answer’s "assess the environmental capacity of this ‘Ecological Museum’, regulate the carrying capacity in terms of population, limit the flow of visitors." Points can be awarded for this. 【Points Awarded】2 points
4. Develop Supporting Facilities: The student proposed constructing supporting facilities and planning them reasonably, which aligns with the standard answer’s "ensure new facilities blend harmoniously with the scenic landscape." Points can be awarded for this. 【Points Awarded】2 points
【Total Points】10 points

B Converted Total Scores for Each Subject

The converted total score for each subject is shown in Table 7. We refer to the proportional weighting of scores for each subject in the GAOKAO over the years.

C Statistics of Question Types

The differences between various question types are significant, as demonstrated by the Table 8:

Subject	Objective Score	Subjective Score	Converted Total Score
Chinese	45	105	150
English	105	45	150
Sciences Mathematics	60	90	150
Humanities Mathematics	60	90	150
Physics	44	66	110
Chemistry	50	50	100
Biology	27	63	90
Politics	50	50	100
History	50	50	100
Geography	40	60	100

Table 7: Converted Total Score for Each Subject.

Question Type	Type	Number	Avg. Length
Biology MCQs	single choice	150	162.1
Chemistry MCQs	single choice	124	385.6
Chinese Lang and Usage MCQs	multi-question choice	56	431.6
Chinese Modern Lit	multi-question choice	29	1872.6
English Cloze Test	five out of seven	26	299.5
English Fill in Blanks	multi-question choice	30	448.3
English MCQs	single choice	105	26.2
English Reading Comp	multi-question choice	124	400.9
Geography MCQs	multi-question choice	34	340.1
History MCQs	single choice	287	158.8
Hum. Math MCQs	single choice	218	157.4
Physics MCQs	multi choice	64	285.5
Political Science MCQs	single choice	320	228.9
Sci. Math MCQs	single choice	214	181.6
Biology Open-ended Questions	open-ended question	116	359.0
Chemistry Open-ended Questions	open-ended question	9	1144.7
Chinese Language Ancient Poetry Reading	open-ended question	29	310.2
Chinese Language Classical Chinese Reading	open-ended question	29	1526.2
Chinese Language Famous Passages and Sentences Dictation	cloze	28	219.5
Chinese Language Language and Writing Skills Open-ended Questions	open-ended question	42	279.4
Chinese Language Literary Text Reading	open-ended question	29	2222.6
Chinese Language Practical Text Reading	open-ended question	24	2069.5
English Language Cloze Passage	cloze	23	198.0
English Language Error Correction	correction	26	120.2
Geography Open-ended Questions	open-ended question	28	173.8
Hum. Math Fill-in-the-Blank	cloze	86	97.7
Hum. Math Open-ended Questions	open-ended question	122	273.6
Physics Open-ended Questions	open-ended question	47	281.5
Political Science Open-ended Questions	open-ended question	60	528.4
Sci. Math Fill-in-the-Blank	cloze	81	112.2
Sci. Math Open-ended Questions	open-ended question	123	273.8

Table 8: **Statistics of Question Types.** Questions above the line are objective questions, while questions below the line are subjective questions.